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United States Patent

12407393

Kind Code

B2

Date of Patent

September 02, 2025

Inventor(s)

Clark; Stephen M. et al.

System and method for spatial awareness overlay onto mobile ad hoc network (MANET) frequent message preambles

Abstract

A communications node of a mobile ad hoc network (MANET) or like multi-node network may receive a preamble and/or header portion associated with a resource allocation message (e.g., as opposed to the full message) transmitted by another network node in motion relative to the receiving node. The receiving node determines a receiver-side Doppler nulling direction (e.g., for offsetting Doppler shift associated with the motion of the transmitting node relative to the receiving node) by adjusting a receiving frequency of the preamble and/or header portion through one or more nulling frequencies, each nulling frequency associated with a nulling direction for offsetting Doppler shift due to relative motion in that direction. Based on the determination of a receiver-side Doppler nulling frequency, the receiving node can determine a velocity and direction of the relative motion between the receiving and transmitting nodes.

Inventors: Clark; Stephen M. (Evansdale, IA), Kwon; Tj T. (Marion, IA), Aanderud; Brian L. (Eden Prairie, MN), Loren; Eric J. (North Liberty, IA), Sorsby; William B. (Cedar Rapids, IA)

Applicant: Rockwell Collins, Inc. (Cedar Rapids, IA)

Family ID: 96876752

Assignee: Rockwell Collins, Inc. (Cedar Rapids, IA)

Appl. No.: 17/957881

Filed: September 30, 2022

Prior Publication Data

Document Identifier

Publication Date

US 20230379033 A1

Nov. 23, 2023

Related U.S. Application Data

continuation parent-doc US 17233107 20210416 US 11726162 child-doc US PCT/US2022/024653
continuation parent-doc US 17541703 20211203 US 12137048 child-doc US 17857920
continuation parent-doc US 17534061 20211123 US 11665658 child-doc US 17857920
continuation-in-part parent-doc US 17857920 20220705 US 12287418 child-doc US 17957881
continuation-in-part parent-doc WO PCT/US2022/024653 20220413 PENDING child-doc US 17857920
continuation-in-part parent-doc US 17408156 20210820 US 11737121 child-doc US 17541703
continuation-in-part parent-doc US 17233107 20210416 US 11726162 child-doc US 17541703
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continuation-in-part parent-doc US 17846625 20220622 PENDING child-doc US 17957881
us-provisional-application US 63344445 20220520

Publication Classification

Int. Cl.: H04B7/06 (20060101); H04L5/00 (20060101)

U.S. Cl.:

CPC H04B7/068 (20130101); H04L5/0048 (20130101);

Field of Classification Search

CPC: H04B (7/068); H04B (7/0667); H04L (5/0048); H04L (5/005); H04L (5/0051)

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Primary Examiner: Ferguson; Keith

Attorney, Agent or Firm: Suiter Swantz IP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application is related to and claims priority from the following U.S. Patent Applications: (a) U.S. patent application Ser. No. 17/233,107, filed Apr. 16, 2021; (b) PCT Patent Application No. PCT/US22/24653, filed Apr. 13, 2022; (c) U.S. patent application Ser. No. 17/408,156, filed Aug. 20, 2021; (d) U.S. patent application Ser. No. 17/534,061, filed Nov. 28, 2021; (e) U.S. patent application Ser. No. 17/541,703, filed Dec. 3, 2021; (f) U.S. patent application Ser. No. 17/846,625, filed Jun. 22, 2022; (g) U.S. patent application Ser. No. 17/857,920, filed Jul. 5, 2022; and (h) U.S. Provisional Patent Application No. 63/344,445, filed May 20, 2022.

(2) Said U.S. and PCT patent applications Ser. Nos. 17/233,107; PCT/US22/24653; 17/408,156; 17/534,061; 17/541,703; 17/846,625; 17/857,920; and 63/344,445 are herein incorporated by reference in their entirety.

BACKGROUND

(3) Mobile Ad-hoc NETworks (MANET; e.g., “mesh networks”) are known in the art as quickly deployable, self-configuring wireless networks with no pre-defined network topology. Each communications node within a MANET is presumed to be able to move freely. Additionally, each communications node within a MANET may be required to forward (relay) data packet traffic.

Data packet routing and delivery within a MANET may depend on a number of factors including, but not limited to, the number of communications nodes within the network, communications node proximity and mobility, power requirements, network bandwidth, user traffic requirements, timing requirements, and the like.

(4) In the context of MANETs and other multi-node communication networks, topology learning is the key task of routing mechanisms especially for proactive routing. Topology learning is non-trivial and requires significant over-the-air control packet exchanges due to the frequent topology changes caused by node mobility and channel condition changes. It is very hard to construct accurate topology of a large MANET. Existing routing protocols utilize small hello packet exchanges to identify neighbor nodes. Hello messaging with neighbor list is a commonly used mechanism to learn the 1.sup.st and the 2.sup.nd hop neighbors. Local topology learned by hello messaging is delivered via packet flooding or hop-by-hop propagation throughout the connected network. By combining local topology information, each and every communication node may identify efficient routes to all connected communication nodes. Whether by hello messaging or packet flooding, however, topology learning and neighbor discovery are major sources of MANET overhead.

(5) In addition to topology factors, fast-moving platforms (e.g., communications nodes moving relative to each other) experience a frequency Doppler shift (e.g., offset) due to the relative radial velocity between each set of nodes. If the Doppler frequency shift is large enough, the communication receiver must implement algorithms that estimate and correct for this doppler shift before performing demodulation. If the receiver does not perform the necessary compensation, the system will operate with a reduced receiver sensitivity and loss of performance—when experiencing large relative velocity between nodes.

SUMMARY

(6) In a first aspect, a receiving (Rx) node of a mobile ad hoc communications network (MANET) or like multi-node network is disclosed. In embodiments, the Rx node receives a preamble and/or header portion (e.g., a full header or partial portion thereof) of a resource allocation message sent by a transmitting (Tx) node of the network, where the Tx node is in motion according to a particular velocity vector (e.g., having a velocity and a direction) relative to a common reference frame known to both nodes (e.g., known to all nodes of the network), and the motion of the Tx node relative to the Rx node is associated with a Doppler frequency shift. The Rx node determines a receiver-side Doppler nulling direction (e.g., to offset the Doppler shift) by adjusting a receiving frequency of the resource allocation message through a set of nulling frequencies each corresponding to a Doppler nulling direction (e.g., where each nulling frequency offsets a relative Doppler shift in the given nulling direction). Based on the frequency adjustment, the Rx node determines a relative velocity and/or relative direction of the Tx node (relative to the Rx node).

(7) In some embodiments, the network is a cell-based network, the Rx node is a participating member of a cell of the network, and the Tx node is an administrator of the cell.

(8) In some embodiments, the Tx node is a tactical base station (TBS) of the network.

(9) In some embodiments, the Rx node is a one-hop neighbor of the Tx node. For example, the Rx node may create a replica message based on the received preamble and/or header portion, e.g., if the resource allocation message is overly noisy or improperly demodulated. The replica message may be correlated against the original resource allocation message, and the replica message used by the Rx node as a basis for adjusting the receiving frequency through the set of nulling frequencies.

(10) In some embodiments, the Rx node is a participating member of a cell, and the Tx node is an administrator or TBS of a different cell of the network.

(11) In some embodiments, the Rx node is in transition from the current cell to the cell administered by the Tx node.

(12) In some embodiments, the common reference frame, as well as the relative motion between the Tx and Rx nodes, is three-dimensional.

- (13) In some embodiments, the Rx node determines the receiver-side nulling direction by adjusting the receiving frequency through a set of step frequencies. For example, each step frequency may correspond to an incremental arc section (e.g., 10, 15, 20 degrees).
- (14) In some embodiments, the Rx node is also in motion according to a receiver-side velocity vector (e.g., having a velocity and a direction). For example, based on receiver-side adjustments to the receiving frequency (e.g., and transmitter-side adjustments to the transmitting frequency by the Tx node), the Rx node determines a relative velocity vector (velocity, direction), e.g., the difference in velocity vectors between the Tx and Rx nodes.
- (15) In some embodiments, the Tx node is associated with a transmitter-side Doppler nulling direction, and the Rx node determines the receiver-side Doppler nulling direction by adjusting the receiving frequency to a nulling frequency corresponding to a nulling direction opposite the transmitter-side Doppler nulling direction.
- (16) In some embodiments, the Rx node is time synchronized with the Tx node.
- (17) In some embodiments, the resource allocation message (or preamble or header portion thereof) is associated with a time of transmission (TOT) by the Tx node, the TOT known to the Rx node. For example, the Rx node determines a time of arrival (TOA) of the resource allocation message/preamble/header portion at the Rx node, and can determine, based on the TOT and TOA, a distance between the Tx and Rx nodes.
- (18) In a further aspect, a transmitting (Tx) node of a MANET or like multi-node network is also disclosed. In embodiments, the Tx node is in motion according to a velocity vector (having a velocity and a direction) relative to a common reference frame known to each node of the network. For example, the Tx node is an administrator or otherwise associated with a cell (e.g., of a cell-based network) and transmits (e.g., at predetermined intervals) to participating nodes of the cell resource allocation messages having a preamble and a header. The Tx node determines a transmitter-side Doppler nulling direction (e.g., to offset Doppler frequency shift associated with the relative motion of the Tx node) by adjusting a transmitting frequency of the preamble or header (or of the resource allocation message) through a nulling frequency corresponding to a particular Doppler nulling direction (e.g., for offsetting Doppler shift due to relative motion in that direction).
- (19) In some embodiments, the Tx node determines the transmitter-side Doppler nulling frequency by adjusting the transmitting frequency through a set of step frequencies, where each step frequency is associated with an incremental arc section (e.g., 10, 15, 20 degrees).
- (20) In some embodiments, the Tx node is a tactical base station (TBS) configured for administration of the network cell.
- (21) In a still further aspect, a method is disclosed. In embodiments, the method includes receiving, via a receiving (Rx) node of a MANET or like multi-node network, a preamble and/or header portion of a resource allocation message sent by a Tx node of the network, the Tx node in motion according to a velocity vector having a velocity and a direction. The method includes determining, via the Rx node, a receiver-side nulling direction (e.g., for offsetting a Doppler frequency shift associated with the motion of the Tx node relative to the Rx node) by adjusting a receiving frequency of the message, preamble, and/or header portion through a nulling frequency corresponding to a Doppler nulling direction (e.g., for offsetting Doppler shift due to relative motion in said direction). The method includes, based on the adjustment of the receiving frequency/determination of the Rx-side nulling frequency, determining a relative velocity vector of the Tx node relative to the Rx node (e.g., a relative velocity, a relative direction).
- (22) In some embodiments, the method includes creating, via the Rx node, a replica message (e.g., a replica header portion and/or replica data payload) based on a noisy or incorrectly demodulated resource allocation message, and performing receiver-side frequency adjustment based on a correlation of the replica message and the resource allocation message.
- (23) In some embodiments, the method includes adjusting, via the Rx node, the receiving frequency through a set of step frequencies, each step frequency corresponding to an incremental

arc section or Doppler nulling direction.

(24) In some embodiments, the Tx and Rx nodes are time synchronized, and the preamble, header portion, and/or message is associated with a time of transmission (TOT) by the Tx node that is also known to the Rx node, and the method includes determining, via the Rx node, a time of arrival (TOA) of the preamble, header portion, and/or message at the Rx node. The method includes determining, based on the TOT and TOA, an estimate of the distance between the Tx and Rx nodes.

(25) In some embodiments, the network is a cell-based network wherein the Rx node is a participating member of a cell within the network, and the Tx node is a tactical base station (TBS) configured for cell administration (e.g., either of the cell of which the Rx node is a participating member, or another cell).

(26) This Summary is provided solely as an introduction to subject matter that is fully described in the Detailed Description and Drawings. The Summary should not be considered to describe essential features nor be used to determine the scope of the Claims. Moreover, it is to be understood that both the foregoing Summary and the following Detailed Description are example and explanatory only and are not necessarily restrictive of the subject matter claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The detailed description is described with reference to the accompanying figures. The use of the same reference numbers in different instances in the description and the figures may indicate similar or identical items. Various embodiments or examples (“examples”) of the present disclosure are disclosed in the following detailed description and the accompanying drawings. The drawings are not necessarily to scale. In general, operations of disclosed processes may be performed in an arbitrary order, unless otherwise provided in the claims. In the drawings:

(2) FIG. 1 is a diagrammatic illustration of a transmitting (Tx) and receiving (Rx) node of multi-node communications network, and the components of each node, according to example embodiments of the inventive concepts disclosed herein;

(3) FIGS. 2A and 2B are plots of net Doppler frequency shift as a function of Doppler nulling angle for a mobile Tx node and stationary Rx nodes of the network of FIG. 1 at various angles;

(4) FIG. 3 is a diagrammatic illustration of the multi-node communications network of FIG. 1 wherein both the Tx and Rx nodes are mobile;

(5) FIGS. 4A and 4B are plots of net Doppler frequency shift as a function of Doppler nulling angle for a mobile Tx and Rx nodes of the network of FIG. 3 at various angles, headings, and/or velocities;

(6) FIG. 5 is a diagrammatic illustration of Doppler null scanning operations of the Tx and Rx nodes of the multi-node communications network of FIGS. 1 and 3;

(7) FIG. 6 is a diagrammatic illustration of a cell-based multi-node communications network according to example embodiments of the inventive concepts disclosed herein;

(8) FIGS. 7A and 7B are diagrammatic illustrations of an operating theater implementation of the multi-node communications network of FIG. 6;

(9) FIGS. 8A and 8B are diagrammatic illustrations of Doppler null overlay operations of the multi-node communications networks of FIGS. 1, 3, and 6 according to example embodiments of the inventive concepts disclosed herein;

(10) and FIG. 9 is a flow diagram illustrating a method for mobile ad hoc networking operations according to example embodiments of the inventive concepts disclosed herein.

DETAILED DESCRIPTION

(11) Before explaining one or more embodiments of the disclosure in detail, it is to be understood that the embodiments are not limited in their application to the details of construction and the

arrangement of the components or steps or methodologies set forth in the following description or illustrated in the drawings. In the following detailed description of embodiments, numerous specific details may be set forth in order to provide a more thorough understanding of the disclosure. However, it will be apparent to one of ordinary skill in the art having the benefit of the instant disclosure that the embodiments disclosed herein may be practiced without some of these specific details. In other instances, well-known features may not be described in detail to avoid unnecessarily complicating the instant disclosure.

(12) As used herein a letter following a reference numeral is intended to reference an embodiment of the feature or element that may be similar, but not necessarily identical, to a previously described element or feature bearing the same reference numeral (e.g., **1**, **1a**, **1b**). Such shorthand notations are used for purposes of convenience only and should not be construed to limit the disclosure in any way unless expressly stated to the contrary.

(13) Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

(14) In addition, use of “a” or “an” may be employed to describe elements and components of embodiments disclosed herein. This is done merely for convenience and “a” and “an” are intended to include “one” or “at least one,” and the singular also includes the plural unless it is obvious that it is meant otherwise.

(15) Finally, as used herein any reference to “one embodiment” or “some embodiments” means that a particular element, feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment disclosed herein. The appearances of the phrase “in some embodiments” in various places in the specification are not necessarily all referring to the same embodiment, and embodiments may include one or more of the features expressly described or inherently present herein, or any combination or sub-combination of two or more such features, along with any other features which may not necessarily be expressly described or inherently present in the instant disclosure.

(16) Referring to FIG. **1**, a multi-node communications network **100** is disclosed. The multi-node communications network **100** may include multiple communications nodes, e.g., a transmitting (Tx) node **102** and a receiving (Rx) node **104**.

(17) In embodiments, the multi-node communications network **100** may include any multi-node communications network known in the art. For example, the multi-node communications network **100** may include a mobile ad-hoc network (MANET) in which the Tx and Rx nodes **102**, **104** (as well as every other communications node within the multi-node communications network) is able to move freely and independently. Similarly, the Tx and Rx nodes **102**, **104** may include any communications node known in the art which may be communicatively coupled. In this regard, the Tx and Rx nodes **102**, **104** may include any communications node known in the art for transmitting/transceiving data packets. For example, the Tx and Rx nodes **102**, **104** may include, but are not limited to, radios, mobile phones, smart phones, tablets, smart watches, laptops, and the like. In embodiments, the Rx node **104** of the multi-node communications network **100** may each include, but are not limited to, a respective controller **106** (e.g., control processor), memory **108**, communication interface **110**, and antenna elements **112**. (In embodiments, all attributes, capabilities, etc. of the Rx node **104** described below may similarly apply to the Tx node **102**, and to any other communication node of the multi-node communication network **100**.)

(18) In embodiments, the controller **106** provides processing functionality for at least the Rx node **104** and can include any number of processors, micro-controllers, circuitry, field programmable gate array (FPGA) or other processing systems, and resident or external memory for storing data, executable code, and other information accessed or generated by the Rx node **104**. For example, the controller **106** may execute one or more software programs embodied in a non-transitory computer

readable medium (e.g., memory **108**) that implement techniques described herein. The controller **106** is not limited by the materials from which it is formed or the processing mechanisms employed therein and, as such, can be implemented via semiconductor(s) and/or transistors (e.g., using electronic integrated circuit (IC) components), and so forth.

(19) In embodiments, the memory **108** can be an example of tangible, computer-readable storage medium that provides storage functionality to store various data and/or program code associated with operation of the Rx node **104** and/or controller **106**, such as software programs and/or code segments, or other data to instruct the controller **106**, and possibly other components of the Rx node **104**, to perform the functionality described herein. For example, the memory **108** can store data, such as a program of instructions for operating the Rx node **104**, including its components (e.g., controller **106**, communication interface **110**, antenna elements **112**, etc.), and so forth. It should be noted that while a single memory **108** is described, a wide variety of types and combinations of memory (e.g., tangible, non-transitory memory) can be employed. The memory **108** can be integral with the controller **106**, can comprise stand-alone memory, or can be a combination of both. Some examples of the memory **108** can include removable and non-removable memory components, such as random-access memory (RAM), read-only memory (ROM), flash memory (e.g., a secure digital (SD) memory card, a mini-SD memory card, and/or a micro-SD memory card), solid-state drive (SSD) memory, magnetic memory, optical memory, universal serial bus (USB) memory devices, hard disk memory, external memory, and so forth.

(20) In embodiments, the communication interface **110** can be operatively configured to communicate with components of the Rx node **104**. For example, the communication interface **110** can be configured to retrieve data from the controller **106** or other devices (e.g., the Tx node **102** and/or other nodes), transmit data for storage in the memory **108**, retrieve data from storage in the memory, and so forth. The communication interface **110** can also be communicatively coupled with the controller **106** to facilitate data transfer between components of the Rx node **104** and the controller **106**. It should be noted that while the communication interface **110** is described as a component of the Rx node **104**, one or more components of the communication interface **110** can be implemented as external components communicatively coupled to the Rx node **104** via a wired and/or wireless connection. The Rx node **104** can also include and/or connect to one or more input/output (I/O) devices. In embodiments, the communication interface **110** includes or is coupled to a transmitter, receiver, transceiver, physical connection interface, or any combination thereof.

(21) It is contemplated herein that the communication interface **110** of the Rx node **104** may be configured to communicatively couple to additional communication interfaces **110** of additional communications nodes (e.g., the Tx node **102**) of the multi-node communications network **100** using any wireless communication techniques known in the art including, but not limited to, GSM, GPRS, CDMA, EV-DO, EDGE, WiMAX, 3G, 4G, 4G LTE, 5G, WiFi protocols, RF, LoRa, and the like.

(22) In embodiments, the antenna elements **112** may include directional or omnidirectional antenna elements capable of being steered or otherwise directed (e.g., via the communications interface **110**) for spatial scanning in a full 360-degree arc (**114**) relative to the Rx node **104**.

(23) In embodiments, one or both of the Tx node **102** and Rx node **104** may be moving in an arbitrary direction at an arbitrary speed, and similarly may be moving relative to each other. For example, the Tx node **102** may be moving relative to the Rx node **104** according to a velocity vector **116**, at a relative velocity  and a relative angular direction (an angle α relative to an arbitrary direction **118** (e.g., due east); **8** may be the angular direction of the Rx node relative to due east.

(24) In embodiments, any signal transmitted by the Tx node **102** and received by the Rx node **104** may be associated with a Doppler frequency shift proportional to the relative radial velocity between the Tx and Rx nodes. For example, the relative radial velocity may be defined by the

motion of the Tx node **102** or of the Tx and Rx nodes **104** (e.g., if both are in motion), in two or three dimensions.

(25) In embodiments, the Tx node **102** may correct for its own Doppler frequency shift (e.g., due to its own motion according to the velocity vector **116**) by implementing a Doppler nulling protocol, e.g., in an earth-centered earth-fixed (ECEF) reference frame (or in some other reference frame as needed). For example, the Tx node **102** may adjust its transmit frequency to counter the Doppler frequency offset such that there is no net frequency offset (e.g., “Doppler null”) for a given Doppler nulling direction **120** (at an angle ϕ relative to the arbitrary direction **118**, e.g., 100 degrees from due east). In embodiments, the communications interface **110** of the Tx node **102** may be informed by the platform (e.g., the controller **106**) of its own velocity vector and orientation (e.g., a, ( custom character) and may adjust its transmitting frequency to remove the Doppler frequency shift at each Doppler nulling direction **120** and angle ϕ . For example, when the Rx node **104** is stationary, the Doppler shift as seen by the Rx node due to the motion of the Tx node **102** may be

$$(26) \frac{f_{DOPPLER}}{f} = \frac{\text{Math.} \cdot \frac{\text{Math.} V_T}{c} \cdot \text{Math.}}{\cos(\text{ } - \text{ })}.$$

In embodiments, the Tx node **102** may adjust its transmitting frequency ($\Delta f_{sub.T}$) to correct for this Doppler shift, based on its speed  and velocity direction α , such that the transmitter frequency adjustment $\Delta f_{sub.T}$ is proportional to the velocity projection onto the Doppler null direction and is

$$(27) \frac{f_T}{f} = \frac{\text{Math.} \cdot \frac{\text{Math.} V_T}{c} \cdot \text{Math.}}{\cos(\text{ } - \text{ })}$$

such that the net frequency shift $\Delta f_{sub.net}$ seen by the Rx node **104** is the sum of these two terms, thus

$$(28) \frac{f_{net}}{f} = \frac{\text{Math.} \cdot \frac{\text{Math.} V_T}{c} \cdot \text{Math.}}{[\cos(\text{ } - \text{ }) - \cos(\text{ } - \text{ })]}.$$

(29) It is assumed that the velocity vector  and velocity direction α will change slowly relative to periodic measurements of $\Delta f_{sub.net}$, such that from the perspective of the Rx node **104** α , , and θ may be regarded as constants.

(30) In embodiments, when the Doppler nulling direction **120** aligns with the receiver direction θ (e.g., the direction of the Rx node **104** relative to the Tx node **102**), the transmitter frequency adjustment $\Delta f_{sub.T}$ may compensate exactly for the net Doppler shift $\Delta f_{sub.net}$. It may further be assumed that the Rx node **104** is configured to resolve any adjustments to the frequency of the inbound transmitted signal.

(31) In embodiments, by plotting (**200**) the net frequency shift as a function of the Doppler nulling direction when the Rx node **104** is stationary and due east of the Tx node **102** ($\theta=0$), and assuming a consistent Tx node velocity and direction (e.g., 1500 m/s, $\alpha=0$ (e.g., due east)) the sinusoidal plot may maintain an amplitude consistent with Tx speed

$$(32) (\frac{|V_T|}{c}) 1e^6,$$

or 5 ppm) regardless of velocity direction or position. Further, net frequency shift is zero when the Doppler nulling angle **120** is aligned with the relative direction of the Rx node **104** (e.g., $\Phi=0$).

Further still, the minimum net frequency shift (e.g., greatest negative frequency shift) occurs when the Doppler nulling direction **120** is aligned with the velocity direction of the Tx node **102** (e.g., $\Phi=\alpha$).

(33) In embodiments, based on the above the stationary Rx node **104** may determine a speed of the Tx node **102** (e.g., , a heading of the Tx node (e.g., α), and a direction of the Tx node relative to the Rx node (e.g., θ). For example, although ambiguities of position direction may exist (e.g., 0 vs. 180 degrees as shown by FIG. 2A; ± 90 degrees as shown by FIG. 2B), it is assumed the Rx node **104** is configured to resolve any such ambiguities, e.g., the Rx node knows whether the Tx node **102** is west or east of it.

(34) Referring now to FIG. 3, the multi-node communications network **300** may be implemented and may function similarly to the multi-node communications network **100** of FIG. 1, except that

the Rx node **302** of the multi-node communications network **300** may also be in motion (e.g., characterized by a receiver velocity V_R and a velocity direction β . In embodiments, the Rx node **302** may adjust its receiver frequency to compensate for its motion as well as the motion of the Tx node **102**. For example, the true Doppler shift as seen by the Rx node **302** (due to the relative radial velocity between the Tx node **102** and the Rx node) may be

$$(35) \frac{f_{DOPPLER}}{f} = \frac{V_T}{c} \cos(\theta - \alpha) - \frac{V_R}{c} \cos(\theta - \beta).$$

(36) In embodiments, both the Tx node **102** and the Rx node **302** may adjust their respective transmitting and receiving frequencies ($\Delta f_{sub.T}$, $\Delta f_{sub.R}$), each according to their own speed (V_T , V_R) and velocity direction (α , β). For example, the Rx node **302** may adjust its receiving frequency either within, or prior to, the frequency resolution algorithm resolving the adjusted transmitting frequency (e.g., $\Delta f_{sub.T}$) of the received signal, thus

$$(37) \frac{f_{T\&R}}{f} = \frac{V_T}{c} \cos(\theta - \alpha) + \frac{V_R}{c} \cos(\theta - \beta).$$

Accordingly, the net frequency shift seen by the Rx node **302** may be the sum of all terms:

$$(38) \frac{f_{net}}{f} = \frac{V_T}{c} [\cos(\theta - \alpha) - \cos(\theta - \beta)] - \frac{V_R}{c} [\cos(\theta - \alpha) - \cos(\theta - \beta)].$$

(39) Referring also to FIGS. **4A** and **4B**, the net frequency shift

$$(40) \frac{f_{net}}{f}$$

for moving Tx and Rx nodes **102**, **302** is plotted (**400**) for various relative receiver angles θ or velocity directions α , β . FIG. **4A** assumes a receiver angle $\theta=0$, while FIG. **4B** assumes common transmitter and receiver velocities ($V_T=V_R$, e.g., 1500 m/s).

(41) Based on FIGS. **4A** and **4B**, it may be observed that amplitude of net frequency shift

$$(42) \frac{f_{net}}{f}$$

(e.g., in ppm) is consistent with relative velocity between the Tx node **102** and Rx node **304**:

$$(43) 0 \leq \frac{V_T \cos \alpha - V_R \cos \beta}{c} \leq 1e^6.$$

Further, as noted above with respect to FIGS. **2A** and **2B**, the net frequency shift is zero when the Doppler null angle **120** aligns with the receiver direction, e.g., $\Phi=0$. Further still, the minimum net frequency shift occurs when the Doppler null angle **120** aligns with the relative velocity direction: $\Phi=(\cos \alpha - \cos \beta)$.

As noted above with respect to FIGS. **2A** and **2B**, the velocity and velocity direction of the Tx node **102** may be known to the Rx node **302**, resolving any dual point ambiguities with respect to the receiver angle θ .

(44) Referring now to FIG. **5**, the multi-node communications network **100** is shown.

(45) In embodiments, the Tx node **102** may scan (**500**) through a full 360 degrees of two-dimensional (2D) space in successive steps or intervals **502** (e.g., 36 cycles of 10 degrees each, 24 cycles of 15°, 18 cycles of 20°) to determine a shift or adjustment to the transmitter frequency (e.g., a step frequency) capable of correcting the net Doppler shift for each interval direction. In some embodiments, less intuitive or more obfuscated scan approaches may be utilized.

(46) In embodiments, both the Tx node **102** and Rx node **104** (as well as the mobile Rx node **302**, FIG. **3**) may implement directional discovery of each other (as well as other nodes of the multi-node communications network **100**) by scanning the full 360 degrees (or a subset thereof). For example, the Tx node **102** and Rx node **104** may be time-synchronized to apply Doppler correction respectively for their own motions relative to a common inertial reference frame. As a transmit angle advances, a receive angle retreats by a same amount as the transmit angle advance. This can be understood by first considering the Tx node **102** when the Tx node applies full Doppler correction in its own direction of travel. Next, consider the Rx node **104** directly in line with the path of travel of the Tx node **102**. If the Rx node **104** at the same time similarly applies Doppler correction for the Rx node's **104** motion in line with the Tx node **102**, then at least a near-zero Doppler path (e.g., a near-zero Doppler path or a zero Doppler path) will exist from the Tx node

102 to the Rx node **104**.

(47) When both the receiver node **104** and the transmitter node apply such synchronized Doppler correction relative to the common inertial reference frame, then the Doppler correction can be swept through a plurality of (e.g., some or all) angles so that a zero Doppler path **504** or near-zero Doppler path will exist from the Tx node **102** to the Rx node **104** including the angle resulting in the near-zero Doppler path or the zero Doppler path, the zero Doppler path having zero net frequency offset. For example, an angle resulting in a near-zero Doppler path may be within 5 degrees of the angle resulting in the zero Doppler path. For any combination of positions and motions of the Tx node **102** and the Rx node **104**, there exists a zero-Doppler path when the Doppler nulling angle **120** is equal to a direction angle ϕ . Accordingly, a zero-Doppler path **504** will be available between the Tx node **102** and the Rx node **104** when the two are synchronized to apply Doppler correction for a swept angle ϕ relative to the inertial reference frame of choice. In embodiments, neither the Tx node **102** nor the Rx node **104** need know a direction to the other node in advance.

(48) In embodiments, when the Doppler nulling direction **120** determined by the Tx node **102** with respect to a given transmitted signal aligns with the Doppler nulling direction **120a** determined by the Rx node **104**, e.g., the two Doppler nulling directions **120**, **120a** are offset by 180 degrees, the Rx node may determine an angle of arrival (AOA) of the transmitted signal, and thus the relative direction/receiver angle θ may be known to the Rx node (e.g., and shared with the Tx node). Further, the Tx node **102** and Rx node **104** may share a time of transmission (TOT) and a time of arrival (TOA; e.g., a time the signal is received by the Rx node) of the transmitted signal, and thereby determine a distance **506** between the Tx and Rx nodes based on the transmission delay. In embodiments, distance estimates between the Tx and Rx nodes **102**, **104** may include an error component proportional to relative bias error between the two nodes. For example, if the Tx and Rx nodes **102**, **104** are not perfectly synchronized, any clock error between the two nodes may produce an error component in the distance estimate (e.g., 1 light-microsecond (300 m $\approx$ 984 ft) for every 1 μ s/1000 ns of clock error).

(49) In some embodiments, the 2D scanning approach may be expanded into 3D space. For example, the 2D scan for a given step direction may correspond to a conical coverage area **508**, e.g., a 10-degree, 15-degree, 20-degree cone, and thereby to a 3D fractional coverage area. Accordingly Doppler null scanning for a full spherical region may be achieved by adding additional scanning cycles corresponding to the areas uncovered by the 2D scan.

(50) Referring now to FIG. **6**, the mobile ad hoc network **600** (MANET) may be implemented similarly to the multi-node communications networks **100**, **300** of FIGS. **1** and **3**, except that the MANET **600** may be organized as a cell-based network of tactical base station (TBS) nodes **602** and user equipment (UE) nodes **604** organized into cells **606**, or subsets of the MANET **600**.

(51) In embodiments, each TBS node **602** and UE node **604** may be implemented and may function similarly to the Tx and/or Rx nodes **102**, **104**, **302** of FIGS. **1** and **3**, except that each cell **606** within the MANET **600** may correspond to a subset of the MANET administered by a TBS node **602**. For example, each cell **606** may correspond to a geographical region or a coverage area of the TBS node **602**.

(52) In embodiments, the MANET **600** may correspond to an operating theater, within which each cell **606** may include a set of participating UE nodes **604** in operation within that cell. For example, within each cell **606** a TBS node **602** may control all network communications and allocate communications resources to all participating UE nodes **604**.

(53) For example, participating UE nodes **604** may include ground-based, water-based, or airborne vehicles and individual manpacks operating either within, or remotely from, a vehicle. Similarly, each TBS node **602** may be a ground-based, water-based, or airborne node (e.g., an unmanned aircraft system (UAS)) in control of all network communications within the cell **606** and configured to allocate communications resources to all participating UE nodes **604** currently

assigned to that TBS node. For example, resource allocations may be transmitted by each TBS node **602** to its assigned participating UE nodes **604** at regular intervals, e.g., each second.

(54) In embodiments, each TBS node **602** and participating UE node **604** may be stationary or mobile. For example, the participating UE node **604a** assigned to the TBS node **602a** and the cell **606a** may be in motion toward the cell **606b** administered by the TBS **602b**. In embodiments, the participating UE node **604a** may be handed off by the TBS node **602a** to the control of the TBS node **602b**, e.g., while operating within the cell **606b**. For example, the participating UE node **604a** may maintain communications with both TBS nodes **602a-b** while preparing for the transition to the cell **606b** and TBS node **602b**. Similarly, the participating UE node **604b**, in motion away from the cell **606b**, may operate within the coverage areas of both TBS nodes **602a-b** but may remain under the administration of the TBS node **602a** and within the cell **606a**.

(55) Referring to FIG. 7A, the multi-node communications network **700** may be implemented and may function similarly to the multi-node communications networks **100**, **300**, **600** except that the multi-node communications network **700** may be implemented in an operating theater.

(56) In embodiments, the multi-node communications network **700** may include cells **606a**, **606b** respectively administered by airborne TBS nodes **702a**, **702b**. For example, the cells **606a**, **606b** may include participating UE nodes embodied in ground-based vehicles **704a-704c** and roaming users **706a-706f** (e.g., manpacks). The roaming users **706a**, **706b** may maintain communications links with the ground-based vehicle **704a** in addition to the airborne TBS **702a**, the roaming users **706c**, **706d** may maintain communications links with the ground-based vehicle **704b**, and the roaming user **706e** may maintain communications links with the ground-based vehicle **704c** (in the latter two cases, in addition to the airborne TBS **702b**).

(57) In embodiments, the roaming user **704f** may be a participating UE node of the cell **606a** assigned to the airborne TBS **702a**, but in motion toward the cell **606b** administered by the airborne TBS **702b** and thereby in communication with both airborne TBS nodes.

(58) Referring also to FIG. 7B, in embodiments the airborne TBS **702a**, **702b** may be capable of long-haul communications links **708** beyond the normal operating range defining the cells **606a**, **606b**. For example, each airborne TBS **702a**, **702b** may receive regular resource allocation messages from the other airborne TBS. Further, the ground-based vehicle **704c** and roaming user **706f** may be within the long-haul range **708** of the airborne TBS **702a** and may receive regular resource allocation messages therefrom.

(59) Referring now to FIG. 8A, a resource allocation message **800** sent by the TBS **602a-602b** of FIG. 6 and the airborne TBS **702a-702b** of FIG. 7 is shown.

(60) In embodiments, the resource allocation message **800** may include, in addition to a data payload **802**, an acquisition preamble **804** and packet header **806**.

(61) Signal acquisition and detection sensitivity in current modern digital communications systems are most often contingent on a digital correlation sequence. Usable length for such a correlation sequence (and consequent resulting sensitivity) may be limited by Doppler shift as the phase increases across the correlator length, resulting in net correlation accumulation amplitude characterized by the sine cardinal (sinc) function and a net correlation energy that goes like the sine cardinal squared (sinc.sup.2) function, e.g.,

(62) $\text{sinc}^2 x = \frac{\sin^2 x}{x^2}$, $x \neq 0$.

(63) With increasing signal frequency offsets, optimal correlator length decreases. Accordingly, the system designer should choose a correlator length suitable to the Doppler requirements. Multiple short length correlation sequences are subsequently often used to allow sensitivity improvement beyond that of a single short correlation sequence (e.g., a random number or pseudorandom number (PR) sequence **808**) but such an approach exhibits degraded sensitivity compared to a single long sequence of the same total length. In the presence of large Doppler shift, even a combination of multiple short sequences may be problematic. While multiple short sequences may be better than nothing, for the same number of bits the approach does not deliver the same

sensitivity attainable with coherent detection of a single long sequence without Doppler shift. In summary, currently, a short correlation sequence is relatively unaffected by Doppler shift but has the drawback of yielding low sensitivity, whereas a long correlation sequence may be capable of yielding high sensitivity but only when Doppler shift is minimal.

(64) In embodiments, Tx nodes (**102**, FIG. 1) and Rx nodes (**104**, FIG. 1; **302**, FIG. 3) may perform frequency adjustments to correct for Doppler shift (e.g., Doppler corrections; Doppler null overlay **810**) over acquisition preambles **804** and/or packet headers **806** transmitted by the Tx nodes and received by the Rx nodes. For example, referring also to FIGS. 7A and 7B above, acquisition preambles **804** and/or packet headers **806** transmitted by TBS nodes **702** (**702a-702b**, FIGS. 7A-B) may be capable of travel and detection by receiving UE nodes (ground-based vehicles **704**, roaming users **706**) throughout the long-haul range (**708**, FIG. 7B) of the transmitting TBS node, whereas payload user data **802** may be reliably detectable only within the cells (**606a**, **606b**; FIG. 7B) corresponding to the normal coverage range of the transmitting TBS node. In some embodiments, acquisition preambles **804** and/or preamble headers **806** may be received and/or detected, in whole or in part, up to three hops away.

(65) In embodiments, transmitting Tx and TBS nodes **102**, **702** may perform Doppler null overlay **810** over the complete acquisition preamble **804** and packet header **806**, while receiving nodes **104**, **302**, **704**, **706** (e.g., N-hop neighbors within the long-haul range **708** of the transmitting TBS node **702**) may perform spatial aware correlations **812** (e.g., adjusting the receiving frequency of the preamble and/or header to correct for Doppler shift based on the motion of the receiving node) on the acquisition preamble and that portion **806a** of the packet header deducible a priori by the receiving node, e.g., based on protocols observed by the receiving node or throughout the multi-node communications network **100**, **300**, **600**, **700** (e.g., including legacy systems). Some examples of this type of a priori header portion **806a** may include node addresses and/or the encoding mode used to modulate the payload. In embodiments, the deduction of the header portion **806a** may be performed by a priori knowledge of which nodes will transmit at a given time, as well as which mode the transmitting nodes will typically use based on recent previous transmissions. For example, if the TBS node **702a** shown by FIGS. 7A and 7B needs to transmit resource allocation messages that can be heard by every participating node (**704a**, **706a**, **706b**, **706f**) within its cell **606a**, the TBS node will have to use the most robust mode to encode its payload data to ensure that the data can be delivered to the furthest nodes (e.g., **706f**). This is an opportunity for the TBS node **102**, **702** (referring back to FIG. 8A) to include the header portion **806a** within the correlation window **812** shown by FIG. 8A.

(66) Referring also to FIG. 8B, one-hop neighbors of the transmitting Tx node **102**/TBS node **702** may receive the full resource allocation message (**800**, FIG. 8A) but in a noisy and/or incorrectly demodulated form **814**. For example, one-hop neighbors may include Rx nodes **104/302**, ground-based vehicles **704**, and/or roaming users **706** at the outer limits of the cell **606a-606b** administered by the Tx node **102**/transmitting TBS node **702**. In embodiments, the resource allocation message **814** may include an intact acquisition preamble **804**, but one or more of the packet header/header portion **806a** and data payload **802a** may be incorrectly modulated due to noise.

(67) In embodiments, the receiving one-hop neighbor may form a replica resource allocation message **816** based on the incorrectly modulated resource allocation message **814**, performing receiver-side passive spatial awareness correlation **818** over the replica resource allocation message. For example, the replica resource allocation message may include a replica packet header **806b** and/or replica data payload **802b**. In embodiments, the receiving one-hop neighbor may construct a replica packet header **806b**, a replica data payload **802b**, or other portions of a replica resource allocation message **816** via error correction coding of the incorrectly modulated data **814**, e.g., via encoding of the received bits to generate redundant parity bits which may be processed by the receiving node to correct erroneously modulated bits. For example, the error-corrected portions of the incorrectly modulated resource allocation message **814** may be remodulated by the receiving

node, producing “corrected” bits for remodulation into the replica packet header **806b** or replica data payload **802b**. In embodiments, the replica resource allocation message **816** may thus serve as a reference for the receiver-side passive spatial awareness correlation **818** against the original incorrectly modulated resource allocation message **814**.

(68) Referring now to FIG. **9**, the method **900** may be implemented by the receiver nodes **104**, **302**, **704**, **706** and may include the following steps.

(69) At a step **902**, the receiving (Rx) node receives at least a preamble or a header portion (e.g., a full header or a portion thereof) of a resource allocation message transmitted by a transmitting (Tx) node within the multi-node communications network of which the receiving node is a participating node, the Tx node in motion according to a velocity and a direction associated with a Doppler shift. In some embodiments, the network may be a cell-based network and the Tx node may be a tactical base station (TBS) configured for administration of a cell of which the receiving node is a participating member (e.g., participating user equipment (UE) such as a vehicle-based node or manpack) or for administration of another cell of the network (e.g., a cell of which the receiving node is not a participating member).

(70) At a step **904**, the receiving node determines a receiver-side Doppler nulling direction capable of correcting for Doppler shift associated with the transmitted resource allocation message and with relative motion of the transmitting Tx node and the receiver node by adjusting a receiving frequency of the resource allocation message through at least one nulling direction. For example, the receiving node may also be in motion and correcting for Doppler shift due to its own velocity and direction. In some embodiments, the receiving node may scan through a range of directions relative to a 2D or 3D inertial reference frame, performing Doppler nulling by adjusting the receiver frequency at one or more interval directions within the range.

(71) At a step **906**, the receiving node determines, based on the receiver-side Doppler nulling and transmitter-side Doppler nulling to correct for Doppler shift due to the motion of the Tx node, one or more of: the relative velocity of the Tx node; the velocity direction of the Tx node; and a direction of the Tx node relative to the receiving node, e.g., based on a determined angle of arrival (AOA) of the received transmission.

(72) At a step **908**, the receiving node may determine a time of arrival (TOA) of the preamble, header portion, and/or resource allocation message.

(73) At a step **910**, the receiving node may determine, based on the determined TOA and a time of transmission TOT of the preamble/header portion/resource allocation message by the Tx node, a distance between the receiving node and the Tx node.

CONCLUSION

(74) It is to be understood that embodiments of the methods disclosed herein may include one or more of the steps described herein. Further, such steps may be carried out in any desired order and two or more of the steps may be carried out simultaneously with one another. Two or more of the steps disclosed herein may be combined in a single step, and in some embodiments, one or more of the steps may be carried out as two or more sub-steps. Further, other steps or sub-steps may be carried in addition to, or as substitutes to one or more of the steps disclosed herein.

(75) Although inventive concepts have been described with reference to the embodiments illustrated in the attached drawing figures, equivalents may be employed and substitutions made herein without departing from the scope of the claims. Components illustrated and described herein are merely examples of a system/device and components that may be used to implement embodiments of the inventive concepts and may be replaced with other devices and components without departing from the scope of the claims. Furthermore, any dimensions, degrees, and/or numerical ranges provided herein are to be understood as non-limiting examples unless otherwise specified in the claims.

Claims

1. A communications node of a multi-node communications network, comprising: a communications interface including at least one antenna element; and a controller operatively coupled to the communications interface, the controller including one or more processors; wherein a common reference frame of the communications network is known to the controller; and wherein the controller is configured to: receive, via the communications interface, at least one of a preamble or a header portion associated with a resource allocation message transmitted by a transmitting (Tx) node of the communications network, the Tx node in motion according to a velocity vector comprising a velocity and a direction; determine a receiver (Rx) nulling direction by adjusting a receiving frequency of at least one of the preamble or the header portion through at least one nulling frequency corresponding to a nulling direction, the Rx nulling direction configured to correct a Doppler frequency shift associated with the velocity vector; and determine, based on the adjusting, one or more of: the velocity of the velocity vector of the Tx node; the direction of the velocity vector of the Tx node; or a direction of the Tx node relative to the communications node.
2. The communications node of claim 1, wherein: the communications node is a participating member of a cell of the communications network; and the Tx node is an administrator of the cell.
3. The communications node of claim 2, wherein the Tx node is a tactical base station (TBS) of the communications network.
4. The communications node of claim 2, wherein: the communications node is a one-hop neighbor of the Tx node; and wherein the communications node is configured to: create a replica message corresponding to the resource allocation message based on the at least one of the preamble or the header portion; and determine the receiver (Rx) nulling direction by adjusting a receiving frequency of the replica message through the at least one nulling frequency.
5. The communications node of claim 1, wherein: the communications node is a participating member of a first cell of the communications network; and the Tx node is an administrator of a second cell of the communications network.
6. The communications node of claim 5, wherein: the communications node is configured to transition from the first cell to the second cell.
7. The communications node of claim 1, wherein the common reference frame is a three-dimensional inertial reference frame.
8. The communications node of claim 1, wherein the communications node is configured to determine the Rx nulling direction by adjusting the receiving frequency through two or more step frequencies, the two or more step frequencies corresponding to an arc section of two or more incremental angles.
9. The communications node of claim 1, wherein the velocity vector is a transmitter (Tx) velocity vector comprising a Tx velocity and a Tx direction, and: the communications node is in motion according to a Rx velocity vector comprising an Rx velocity and an Rx direction; and wherein the controller is configured to determine, based on the adjusting, one or more of: a relative velocity of the Tx node relative to the communications node; and a relative direction of the Tx node relative to the communications node.
10. The communications node of claim 9, wherein: the Tx node is associated with a Tx nulling direction; and wherein the controller is configured to determine the Rx nulling direction by adjusting the receiving frequency of at least one of the preamble or the header portion to an Rx nulling direction opposite the Tx nulling direction.
11. The communications node of claim 9, wherein the controller is configured to determine the Rx nulling direction based on a time synchronization with the Tx node.
12. The communications node of claim 11, wherein: at least one of the preamble, the header portion, or the resource allocation message is associated with a time of transmission TOT known to

the controller; and wherein the controller is configured to: determine a time of arrival (TOA) associated with at least one of the preamble, the header portion, or the resource allocation message; and based on at least the TOA and the TOT determine a distance between the communications node and the Tx node.

13. A communications node of a multi-node communications network, comprising: a communications interface including at least one antenna element; and a controller operatively coupled to the communications interface, the controller including one or more processors; wherein a common reference frame of the communications network is known to the controller, wherein the communications node is in motion according to a velocity vector comprising a velocity and a direction; and wherein the communications node is a transmitting (Tx) node associated with a cell of one or more participating communications nodes; wherein the controller is configured to: transmit, via the communications interface, at least one resource allocation message to the one or more participating communications nodes, each resource allocation message including at least one of a preamble or a header portion; and determine a transmitter (Tx) nulling direction by adjusting a transmitting frequency of at least one of the preamble or the header through at least one nulling frequency corresponding to a nulling direction, the Tx nulling direction configured to correct a Doppler frequency shift associated with the velocity vector.

14. The communications node of claim 13, wherein the communications node is configured to determine the Tx nulling direction by adjusting the transmitting frequency through two or more step frequencies, the two or more step frequencies corresponding to an arc section of two or more incremental angles.

15. The communications node of claim 13, wherein the Tx node is a tactical base station (TBS) configured for administration of the cell.

16. A method, comprising: receiving, via a communications node of a multi-node communications network, at least one of a preamble or a header portion associated with a resource allocation message transmitted by a transmitting (Tx) node of the communications network, the Tx node in motion according to a velocity vector comprising a velocity and a direction; determining, via the communications node, a receiver (Rx) nulling direction by adjusting a receiving frequency of at least one of the preamble or the header portion through at least one nulling frequency corresponding to an Rx nulling direction, the Rx nulling direction configured to correct a Doppler frequency shift associated with the velocity vector; and determining, via the communications node and based on the adjusting, one or more of: the velocity of the velocity vector of the Tx node; the direction of the velocity vector of the Tx node; or a direction of the Tx node relative to the communications node.

17. The method of claim 16, wherein determining, via the communications node, a receiver (Rx) nulling direction by adjusting a receiving frequency of at least one of the preamble or the header portion through at least one nulling frequency includes: creating, based on at least one of the preamble or the header portion, a replica message corresponding to the resource allocation message; and determining the Rx nulling direction by adjusting a receiving frequency of the replica message through the at least one nulling frequency.

18. The method of claim 16, wherein determining, via the communications node, a receiver (Rx) nulling direction by adjusting a receiving frequency of at least one of the preamble or the header portion through at least one nulling frequency includes: adjusting the receiving frequency through two or more step frequencies, the two or more step frequencies corresponding to an arc section of two or more incremental angles.

19. The method of claim 16, wherein: the communications node and the Tx node are time synchronized; wherein at least one of the preamble, the header portion, or the resource allocation message is associated with a time of transmission TOT known to the communications node; further comprising: determining, via the communications node, a time of arrival (TOA) associated with at least one of the preamble, the header portion, or the resource allocation message; and determining,

via the communications node and based on at least the TOA and the TOT a distance between the communications node and the Tx node.

20. The method of claim 16, wherein: the multi-node communications network is a cell-based network comprising a plurality of cells; the communications node is a participating member of a first cell of the plurality of cells; and the Tx node is a tactical base station (TBS) configured for administration of at least one cell of the plurality of cells.
